

Scheme 5

Psychological Concept Clustering and Framework Mapping Scheme

Prompt-based normalization of detected psychological concepts

Purpose

This scheme standardizes higher-order concept mapping after concept detection. It groups extracted psychological concepts from chronic pain review records into interpretable families and links them to likely theoretical frameworks. The goal is to support cross-record comparability when the raw concepts are heterogeneous, overlapping, or variably named.

Operational Status

- Workflow position: post-detection concept normalization.
- Operational mode: fixed pattern-based concept detection upstream, followed by LLM clustering for higher-order normalization.
- Canonical implementation basis: `src/bps_review/extraction/coding.py`.

Canonical Source Paths

- `src/bps_review/extraction/coding.py`
- `src/protocol/codebooks/stage2_codebook.md`
- `src/protocol/codebooks/stage3_codebook.md`
- `src/data/interim/extraction/llm_concept_clusters.json`

Upstream Detection Seeds

Psychological Concept Patterns

- fear-avoidance
- catastrophizing
- kinesiophobia
- depression
- anxiety
- coping
- self-efficacy
- acceptance

- illness perceptions
- pain beliefs
- distress
- expectations
- emotion regulation
- mindfulness
- trauma
- sleep

Framework Seeds

- fear-avoidance model
- cognitive behavioral therapy
- acceptance and commitment therapy
- illness perception framework
- operant learning
- social cognitive theory
- self-regulation

Prompt Specification

The implemented system prompt is a strict JSON instruction with the following functional specification:

- Cluster psychological chronic pain review concepts into higher-order families.
- Note likely theoretical frameworks.
- Return strict JSON with a top-level `clusters` list.
- Each cluster must contain `family`, `members`, and `possible_frameworks`.

Expected Output Structure

clusters	Top-level list of normalized concept clusters.
family	Higher-order concept family label assigned to the cluster.
members	The original concept strings grouped into that family.
possible_frameworks	Likely theoretical or therapeutic frameworks associated with the family.

Analytic Role

- Preserves the original detected concepts while improving comparability across records.
- Supports synthesis questions about which psychological concepts and frameworks dominate the review corpus.
- Allows concept families to be interpreted at a higher level than raw string matching alone.

Interpretive Note

This scheme is distinct from the Stage 2 and Stage 3 codebooks because it does not classify whole records. Instead, it normalizes the concept vocabulary itself and therefore functions as a second-order coding scheme over concept strings extracted from the review texts.